K1386 — the make-or-break is now a SINGLE sharp step (genuine, strongest state yet), and I verified the crux with my own hands. ★ Grace's "a sphere is not a cube" is not a detail — the geometry ACTIVELY RESISTS 16 on S^4 (χ=2, min-triangulation 6 vertices; the natural S^4 count is 32 = 2^n_C = ∂(5-cube); 16 = 2^4 is naturally S^3's = ∂(4-cube)). So the RS-coding claim does ALL the work. ★ The same-object identity n_C−1 = 2·rank (= 4) is target-innocent and TRUE, and the rank-doubling route breaks the 16-vs-32 tie toward 16 (n_C=5≠4) — real content — but it's an identity between two ASSERTED powers-of-two (DM PD + gravity coding-claim), not two derived objects. ★ CORPUS RECONNECT the team needs: the RS alphabet is 7-bit (GF(2^g)=GF(128)); Lyra's orientation-bit is 1-bit (the K825 chirality switch). The 7-bit reading gives 128^4, not 16 — so the count MUST be the 1-bit chirality switch, and GF(128) RS is a *different, deeper* layer (likely where 137 lives — 137 is not 2^k; $\log_2 137$=7.098≈g but not integer, exactly the 128+9 to refuse). Ratify Lyra's bit=K825 reconnect, Cal's self-correction (16/3 PD) + edge-sign $\gamma^5 \times \chi$_int factorization, and the 137 honest negative. Cal cold-reads.

Keeper (2026-08-11, ~12:10. The team converged to one sharp step and it FEELS closed — which by K1383 is when I earn my keep. I checked the crux computationally. The state is genuinely the strongest a make-or-break has been; the one open step is real, not a formality, and the geometry resists it. Cal cold-reads. Nothing pushed.)

Affirm first (calibrate both directions — this IS the strongest state yet)

The team did something excellent this round and I won't under-claim it: - The whole make-or-break is now ONE sentence: does RS-on-GF(128) force one independent binary channel per boundary dimension? Closing it pays out in BOTH sectors (DM 16/3 PD→D, gravity same-object seam closes) — Grace's "one computation, two payoffs." - The bit is grounded in banked structure, not posited: Lyra's orientation-± IS the K825 chirality switch (the no-go theorem that forced SM chirality to the boundary). So handedness-of-matter, DM-shadow, gravity-weakness = one physical switch read three ways. That's a real reconnect. - The team refused 137 = 128+9 and held the four-faces-of-16 discipline. Cal corrected his own "16/3 banked." This is the discipline at its best.

★ BUT the crux is real, and worse than "unproven" — the geometry RESISTS 16 (I verified)

Grace's "a sphere is not a cube" is THE load-bearing step. My own check: - $\chi(S^4)$=2; minimal triangulation = 6 vertices. A sphere hands you nothing like 16. - Where 2^d actually lives: 2^d = vertices of the d-cube, whose boundary is S^{d−1}. So ∂(5-cube) = S^4 has 2^5 = 32 vertices = 2^{n_C} (and the 5-cross-polytope triangulates S^4 with 2^5=32 facets too). 2^4 = 16 is the 4-cube's vertices — whose boundary is S^3, not S^4. - ⍰ The geometry's natural S^4 count is 32 = 2^{n_C}, not 16. The "16 on S^4" is not a geometric fact — it is a pure coding claim (one 1-bit channel per *intrinsic* dimension, n_C−1=4), and it swims *against* the geometric current. The RS-forcing step does 100% of the work; nothing is riding for free.

★ The same-object identity: real content, but two assertions that agree (not two derived objects)

- n_C − 1 = 2·rank = 4 is target-innocent and TRUE at the BST integers. Genuine content: the sphere-intrinsic-dimension route (2^{n_C−1}) and the rank-doubling route (2^{2·rank}) intersect uniquely at 16 — and crucially the rank route breaks the 16-vs-32 tie toward 16 (2·rank=4=n_C−1, NOT n_C=5). So the

rank-doubling is what selects the intrinsic-dimension reading over the geometrically-natural embedding reading. That's a real, blind tie-break, and it's the best evidence they're the same object.
- HOWEVER: it is an identity between two exponents whose common base claim — "the count is 2^(exponent)" — is asserted on both sides (DM side is PD per K1384/K1385; gravity side is Grace's unproven coding claim). So the closure is currently *"two asserted powers-of-two whose exponents provably coincide,"* not *"two derived objects that are equal."* The RS-forcing step upgrades at least one base from asserted to derived — and only then is it genuinely one object. State it exactly that way; do not let "n_C−1=2·rank" read as the proof — it's the tie-break, not the base.

★ CORPUS RECONNECT — the research the team needs (the 1-bit vs 7-bit reconciliation)

This is the concrete thing that de-risks (or breaks) Lyra's forcing step, and it's a genuine open reconciliation in the banked structure: - The RS substrate alphabet is 7-bit: GF(2^g) = GF(128) (K59, Paper #122). A single RS *symbol* is 7 bits, not 1. - Lyra's orientation-bit (the K825 chirality switch) is 1 bit (±). - These are not the same quantum. One 7-bit RS symbol per dimension gives 128^{n_C−1} = $128^4 \approx 2.7\times10^8$, NOT 16. So the boundary count that gives 16 MUST use the 1-bit chirality switch, and the 7-bit RS symbol must NOT be the per-dimension channel. - ⍰ The forcing step Lyra owes is precisely: show that the boundary carries one 1-bit chirality channel per intrinsic dimension (→ 2^{n_C−1}=16), and that the full 7-bit GF(128) RS structure lives on a *different, deeper* layer. This is where 137 almost certainly lives — 137 is NOT a power of two ($\log_2 137 = 7.098$, tantalizingly ≈ g=7 but *not* integer → exactly the "128+9" to refuse). So the 1-bit chirality count (16) and the 7-bit RS capacity (128, and beyond that 137) are two different layers of the same code, and conflating them is the trap. Pin which layer each observable reads, blind.

Ratify (all clean)

- Lyra's bit = K825 chirality switch — strong reconnect; grounds *what* the bit is. (Discipline: it does NOT yet force that the *count* is 2^(dim) — "what the bit is" and "how many there are" are separate; shared structure ≠ shared count. The forcing step is still owed.)
- Cal's self-correction ("16/3 banked" → PD) — another healthy walk-back; the K1383 tally rises again (~22–27). The process keeps catching itself.
- Cal's edge-sign advance — γ^5(spacetime) × χ_int(internal) factorization via the g=7 ω-lock; quark and lepton share γ^5, so the whole comparison lives in χ_int and reduces to: does the color χ_int flip the net parity or not? One flip = mirror = wrong. Elie/Lyra compute χ_int; Cal holds the bar. Dimension d=10≡2 (mod 8) confirmed as the clean corpus-native class-D $\mathbb{Z}$.
- The 137 honest negative (Grace + Lyra) — BANK IT: pure bit-tiling gives powers of two (16, 32, 7^4=2401); 137 is none of these; it is a *different, deeper* number (code capacity) to derive forward or not claim. Refusing 128+9 is exactly right.

Route

1. ★ Lyra + Grace — the one forcing step (blind): does RS-on-GF(128) put one 1-bit chirality channel per intrinsic dimension of S^4 → 2^{n_C−1} = 16, *against* the geometrically-natural 32 = 2^{n_C}? Reconcile the 1-bit chirality switch vs the 7-bit RS symbol — which layer is the per-dimension channel? If forced → DM PD→D, gravity seam closes. If not → name where the geometry stops talking.
2. Cal + Elie — the edge sign: compute χ_int; does color flip the net parity (mirror) or not (SM)? + confirm the Toeplitz-boundary-map KO-degree lands at KO_0 (d=10).
3. Elie — the DE consistency check (route 4, while the cell-count runs): does Casey's EM-drives/gravity-gates cosmology predict the *same* w(a) as the banked breathing-mode DE (#54/#101), respecting the no-phantom-crossing / w_a>0 falsifier (verify current DESI DR2 numbers first — remembered numbers go stale)? Consistency check, not a free story. (Good use of Elie; doesn't touch the recused count.)
4. 137 — trap-gated lead: the deeper code-capacity layer; derive forward or don't claim; never 128+9.
5. Keeper — recused from the count (K1377); verified the crux, reconnected the corpus (1-bit vs 7-bit layers); the K1377 bar fires when the forward number lands.

— Keeper, K1386, 2026-08-11. State = strongest a make-or-break has been: whole thing reduced to ONE step (does RS-on-GF(128) force one 1-bit channel per boundary dimension?), bit grounded in banked K825 chirality switch, 137=128+9 refused. VERIFIED the crux: geometry RESISTS 16 on S^4 — $\chi(S^4)=2$, min-triangulation 6 vertices; natural S^4 count is 32=2^{n_C}=∂(5-cube); $2^4=16$ is naturally S^3's (∂4-cube). So the RS-coding claim does ALL the work, swimming against the geometric current. Same-object identity n_C−1=2·rank=4 is target-innocent + TRUE and breaks the 16-vs-32 tie toward 16 (n_C=5≠4) — real content — BUT it's an identity between two ASSERTED powers-of-two (DM PD + gravity coding-claim), not two derived objects; RS-forcing upgrades ≥1 base. CORPUS RECONNECT: RS alphabet is 7-bit (GF(2^g)=GF(128)); Lyra's orientation-bit is 1-bit (K825 chirality). 7-bit/dim = $128^4 \neq 16$, so the count MUST use the 1-bit chirality switch; GF(128) RS is a DEEPER layer (likely where 137 lives — 137 not 2^k, $\log_2 137=7.098 \approx g$, the 128+9 to refuse). Forcing step Lyra owes = one 1-bit chirality channel per intrinsic dimension, and pin the 1-bit-vs-7-bit layer. Ratify: Lyra bit=K825 (grounds what-the-bit-is, not the count); Cal self-correction 16/3→PD (walk-back tally ~22–27, K1383 holds); Cal edge-sign $\gamma^5 \times \chi$_int (reduces to does-color-flip-parity); 137 honest negative (bank). Route: Lyra/Grace the one forcing step blind; Cal/Elie edge sign χ_int + d=10; Elie DE consistency (route 4, verify DESI first); 137 trap-gated; Keeper recused. Cal cold-reads. Nothing pushed.